Bit Error Rate Analysis and BER-Constrained Power Allocation for Downlink Multirate NOMA with Low-Complexity MR-SIC Detection

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***Abstract*—Multirate non-orthogonal multiple access (MR-NOMA) adds the symbol duration to the usual power-domain design variables, so that the symbol energy of every user is set jointly by its power coefficient and its symbol rate. This paper studies the downlink two-user MR-NOMA system with arbitrary square-QAM modulation orders and symbol durations. We implement the joint-multiuser maximum-likelihood sequence detector (JMLSD) through a pairwise-error-probability union bound, together with a low-complexity multirate successive interference cancellation (MR-SIC) receiver whose far-user and near-user bit error rates (BERs) are evaluated in closed form. Monte Carlo simulation confirms the closed-form expressions, and shows that MR-SIC and JMLSD coincide at moderate and high signal-to-noise ratio (SNR) while the detector search space shrinks from $M_1^{\tau_1}M_2$ sequences to a few single-user decisions. Doubling the far-user symbol duration gives a 3 dB energy gain plus a weak-interference cancellation gain that grows with the near-user modulation order; at a BER of 10^{-2} the total gain at $T_2 = 8$ reaches 13.2 dB for BPSK/QPSK and 21.6 dB for [64,4]. Finally, two BER-constrained power-allocation problems are solved numerically for [4,4], [16,4] and [64,4], showing that the optimal near-user power share rises with T_2 until it saturates at its reliable-SIC limit, and quantifying the rate that is traded for reliability.**

***Index Terms*—Bit error rate (BER), joint-multiuser maximum likelihood sequence detector (JMLSD), multirate communications, non-orthogonal multiple access (NOMA), power allocation, successive interference cancellation (SIC).**

I. INTRODUCTION

Non-orthogonal multiple access (NOMA) lets several users share one time–frequency resource by superposing their signals in the power domain. It was first proposed for cellular systems to raise spectral efficiency [1] and has since become a leading candidate for 5G and beyond [2], [3], [4]. Surveys of power-domain NOMA discuss its potential, its open challenges and its integration into LTE and 5G standards [5], [6]. Each receiver separates the superposed signals by successive interference cancellation (SIC), so the quality of detection depends strongly on how the transmit power is divided among the users [7].

Early performance studies of NOMA concentrated on outage probability and capacity, for randomly deployed users [8], for different user pairings [9] and for cooperative transmission [10]. The bit error rate (BER), which is the natural metric for uncoded and low-latency links, was derived exactly for arbitrary numbers of users and modulation orders in [11], and adaptive modulation and power under block-error constraints were studied in [12]. An error in the stronger interfering signal propagates to the weaker one, so power allocation must also guarantee reliable SIC [13]. Joint design of power allocation and decoding order has been treated for outage-optimal operation [14], cognitive NOMA with unknown transmission modes in [15], and user grouping in [16].

Most of these error-rate studies consider single-rate NOMA (SR-NOMA), where all users share one symbol rate. The design freedom is then limited to the modulation orders and the power coefficients, which restricts the reliability targets that can be met simultaneously. Multirate NOMA (MR-NOMA) adds a third variable: the users are given different symbol durations, so the energy a user collects per symbol is the product of its power coefficient and its symbol period [17]. A far user can be given a long symbol period to gain energy without taking a larger share of the transmit power, while the near user keeps a short symbol period and a high data rate. This matches the heterogeneous service classes of future networks, where delay-sensitive devices need short symbols and delay-tolerant devices can trade rate for reliability.

Other ways of introducing flexible overlap include multinumerology NOMA, which mixes subcarrier spacings and is limited by inter-numerology interference [18], [19], [20], [21], and partial NOMA, which tunes the frequency overlap between users [22], [23], [24]. MR-NOMA instead targets single-carrier systems and uses the symbol duration as the source of diversity. Numerical BER results without analysis were first reported in [25], [17], and the closed-form BER analysis and the low-complexity detector for the downlink were derived in [26].

This paper builds on the MR-NOMA framework of [26] and its public MATLAB implementation [27]. We re-implement the analytical expressions, validate them by Monte Carlo simulation, and use them to study the design trade-offs. The contributions are:

1) The JMLSD union bound and the MR-SIC BER expressions for the far and near users are implemented for arbitrary square-QAM orders, validated by simulation, and the two detectors are compared over SNR and over the power coefficient α_1.

2) The far-user BER is evaluated for ten modulation pairs from [2,2] to [64,64] and for $T_2 \in \{1, 2, 4, 8\}$, and the SNR needed for a BER of 10^{-2} is tabulated to separate the processing gain from the weak-interference cancellation gain.

3) The detection complexity of JMLSD and MR-SIC is counted for the [4,4] system.

4) The two BER-constrained power-allocation problems are solved for [4,4], [16,4] and [64,4] using the perfect-SIC lower bound for the near-user BER, and the optimal α_1^*, the resulting BERs and the sum rate are reported versus T_2.

The rest of the paper is organized as follows. Section II presents the system model and the two detectors. Section III derives the JMLSD bound, and Section IV the MR-SIC BER. Section V formulates the power allocation, Section VI discusses the numerical results, and Section VII concludes. Appendices describe the simulation procedure and numerical aspects of the implementation.

II. SYSTEM MODEL

A. Signal Model

We consider a downlink in which a base station (BS) serves N users; $N = 2$ is used in the numerical results. User U_n has a normalized symbol duration T_n with $T_1 < T_2 < \cdots < T_N$, so U_1 is the near, highest-rate user and U_N the far, lowest-rate user. All receivers sample at the rate of U_1, which requires $\tau_n = T_N/T_n$ to be an integer. We set $T_1 = 1$, so $\tau_1 = T_2$ for two users. Over a window of T_N samples the BS transmits

$$x[\ell] = \sum_{n=1}^{N} \sqrt{\alpha_n}\, x_n[\ell], \quad \ell \in \{1, \ldots, T_N\} \tag{1}$$

where $x_n[\ell]$ is an M_n-ary square-QAM symbol with unit average energy, α_n is the power coefficient, $\sum \alpha_n = 1$, and $\alpha_1 < \alpha_2 < \cdots < \alpha_N$ so that the far user receives most of the power. Within the window U_n sends τ_n independent symbols, each held for T_n consecutive samples. Stacking the samples gives

$$\mathbf{x} = \sum_{n=1}^{N} \sqrt{\alpha_n}\, \mathbf{x}_n \tag{2}$$

For $T_2 = 2$, for instance, the two samples carry the same far-user symbol x_2 but two independent near-user symbols, $x[\ell] = \sqrt{\alpha_1}\, x_1[\ell] + \sqrt{\alpha_2}\, x_2$, $\ell = 1, 2$ (Fig. 1). The key symbols used in the paper are collected in Table I.

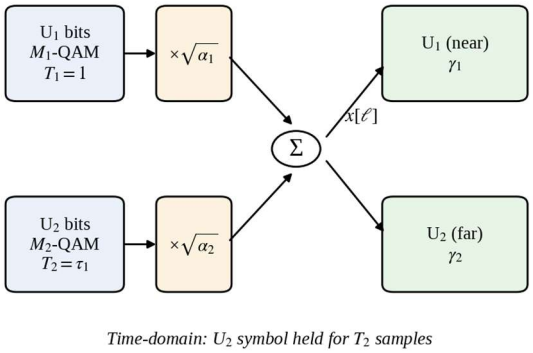


Fig. 1. Downlink MR-NOMA with two users. U_2 (far) holds each symbol for $T_2 = \tau_1$ samples, while U_1 (near) sends a new symbol in every sample.

TABLE I
Main symbols

Symbol	Meaning
N, M_n	Number of users; modulation order of U_n
T_n, τ_n	Symbol duration of U_n; $\tau_n = T_N/T_n$
α_n, $\alpha_{1,\max}$	Power coefficient; reliable-SIC limit (4)
κ_n, Δ_n	QAM energy factor and amplitude index (3)
γ_n, λ_n	SNR of U_n; $\lambda_n = \sqrt{\gamma_n}$ (20)
$\tilde{\lambda}_n$	Effective $\lambda_n/\sqrt{\tau_1}$ after sample combining (26)
P_{Bn}	BER of U_n
ζ	BER threshold of the optimization

For square QAM the energy normalization and the maximum in-phase amplitude index are

$$\kappa_n = \frac{2(M_n-1)}{3}, \quad \Delta_n = \sqrt{M_n} - 1 \tag{3}$$

SIC is reliable only if α_1 stays below [13]

$$\alpha_{1,\max} = \frac{\kappa_1}{\kappa_1 + \kappa_2 \Delta_1^2} \tag{4}$$

which equals 0.5, 0.357 and 0.3 for [4,4], [16,4] and [64,4]. Unless stated otherwise, the equal-minimum-distance allocation [12]

$$\alpha_1 = \frac{M_1 - 1}{M_1 M_2 - 1} \tag{5}$$

is used. It turns the superposed constellation into a uniform square grid (Fig. 3), in which the minimum distance between neighboring points is the same for the near-user and far-user decision boundaries. The channel is AWGN with $h_n = 1$, SNR $= 1/N_0$, and the received sequence of U_n is

$$\mathbf{r}_n = h_n \mathbf{x} + \mathbf{w}_n \tag{6}$$

with $\mathbf{w}_n \sim \mathcal{CN}(0, N_0\mathbf{I})$.

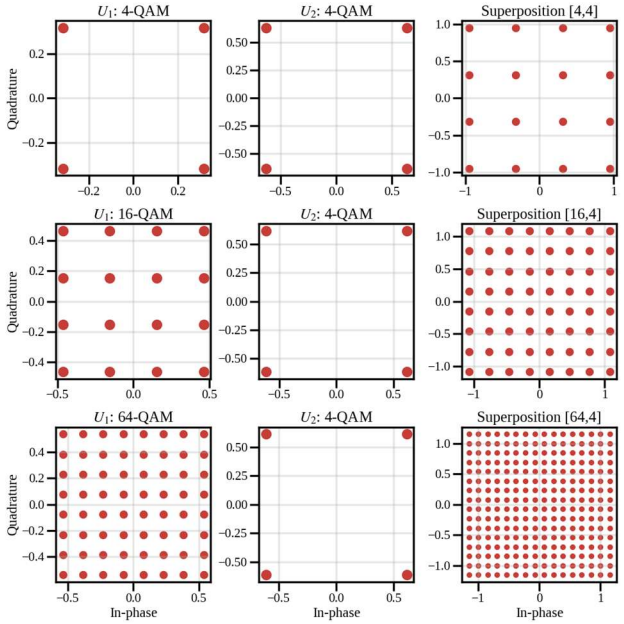


Fig. 3. Constellations of U_1 (left, scaled by $\sqrt{\alpha_1}$), U_2 (middle, scaled by $\sqrt{\alpha_2}$) and the superposed NOMA constellation (right) for [4,4], [16,4] and [64,4] with the equal-distance allocation (5).

B. Optimal Detector: JMLSD

Since the noise is white and Gaussian, the maximum-likelihood detector of the whole window is the minimum-distance rule of classical multiuser detection [28]:

$$\{\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2\} = \text{argmin}_{\mathbf{x}} \parallel \mathbf{r}_n - h_n \mathbf{x} \parallel_2^2 \tag{7}$$

with $\mathbf{x}$ ranging over all $M_1^{\tau_1} M_2$ candidate sequences. For equal symbol rates it reduces to the symbol-by-symbol joint maximum-likelihood detector.

C. Low-Complexity MR-SIC Detector

Because $\alpha_N > \alpha_n$ and $T_N > T_n$ for $n < N$, the detection order is $x_N, x_{N-1}, \ldots, x_1$ (Fig. 2). The far user sums the T_N samples, which collects the energy of its repeated symbol, and applies a single-user maximum-likelihood decision (MLD):

$$\hat{x}_N = \text{argmin}_{x_N} |\mathbf{1}^T \mathbf{r}_n - h_n \sqrt{\alpha_N}\, T_N\, x_N|^2 \tag{8}$$

A user with $n < N$ reconstructs the already detected, higher-energy signals and subtracts them,

$$\tilde{\mathbf{r}}_n = \mathbf{r}_n - h_n \sum_{i=n+1}^{N} \sqrt{\alpha_i}\, \hat{\mathbf{x}}_i \tag{9}$$

then splits the residual into τ_n vectors of length T_n and detects each symbol with a single-user MLD,

$$\hat{x}_n = \text{argmin}_{x_n} \left| \mathbf{1}^T \tilde{\mathbf{r}}_n^{\,k} - h_n \sqrt{\alpha_n}\, T_n\, x_n \right|^2 \tag{10}$$

with $k = \lceil \ell / T_n \rceil$ and $\tilde{\mathbf{r}}_n^{\,k}$ the k-th partition. For $T_n = 1$ the receiver becomes the conventional SR-SIC. All users must know the parameters α_i, T_i and M_i of the users with higher energy than themselves, which the BS can broadcast; the signaling overhead is therefore similar to that of SR-NOMA. MR-SIC and JMLSD are equivalent [26], and our simulations in Section VI confirm that they produce the same BER; the difference between the curves comes only from the union bound used for JMLSD.

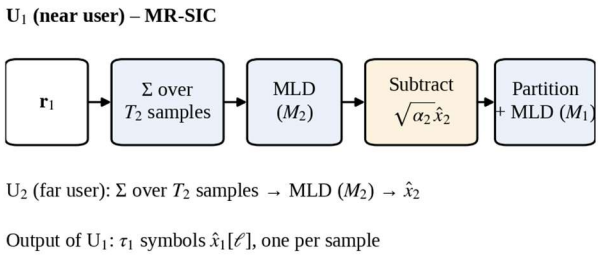


Fig. 2. MR-SIC detection for the near user U_1; the far user stops after the first MLD.

D. Complexity

The JMLSD search space has size

$$|\mathcal{X}| = \prod_{i=1}^{N} M_i^{\tau_i} \tag{11}$$

Counting real multiplications (RM) and additions (RA) per candidate gives

$$C_{\text{JMLSD}}^{\text{RM}} = (12 + 10T_N)\,|\mathcal{X}| \tag{12}$$

$$C_{\text{JMLSD}}^{\text{RA}} = (11 + 7T_N)\,|\mathcal{X}| \tag{13}$$

whereas MR-SIC needs only single-user decisions for U_n, so that

$$C_{\text{SIC}}^{\text{RM}} = (N-n)(4T_N + 2) + \sum_{i=n}^{N} (3 + 8M_i)\,\tau_i \tag{14}$$

$$C_{\text{SIC}}^{\text{RA}} = (N-n)(5T_N + 1) + \sum_{i=n}^{N} (2T_i + 6M_i + 3)\,\tau_i \tag{15}$$

grow linearly in M_i and τ_i. Table II evaluates the RM counts for [4,4] and $n = 1$: at $T_2 = 8$ the JMLSD must test $4^9 = 262\,144$ sequences, and MR-SIC is about 7×10^4 times cheaper. The RA counts show the same behavior. MR-SIC does require buffering of τ_1 near-user samples before detection starts, which adds a small delay, and sampling at the symbol rate of U_1, which makes timing synchronization more demanding for the low-rate users.

TABLE II
Real-multiplication count, [4,4], $n = 1$

T_2	$\lvert\mathcal{X}\rvert$	JMLSD	MR-SIC	Ratio
1	16	352	76	4.632
2	64	2 048	115	17.809
4	1 024	5.32×10^4	193	275.896
8	2.62×10^5	2.41×10^7	349	6.91×10^4

III. BER ANALYSIS OF THE JMLSD

For coherent detection the channel phase θ_n is removed, giving $\check{\mathbf{r}}_n = |h_n|\mathbf{x} + e^{-*j\theta*_{-\{n\}}}\mathbf{w}_n$, and the detector minimizes the metric

$$\phi_n(\bar{\mathbf{x}}) = \parallel \check{\mathbf{r}}_n - |h_n|\bar{\mathbf{x}} \parallel_2^2 \tag{16}$$

over the trial sequences $\bar{\mathbf{x}}$. An exact BER is impractical because of the number of sequence pairs, so the pairwise error probability (PEP) is used. The sequence $\mathbf{x}$ is mistaken for $\bar{\mathbf{x}}$ when

$$\Pr(\mathbf{x} \to \bar{\mathbf{x}}) = \Pr(\text{Re}\{(\mathbf{x} - \bar{\mathbf{x}})^H \mathbf{w}_n\} > 1/2\,|h_n|\ \parallel \mathbf{x} - \bar{\mathbf{x}} \parallel_2^2) \tag{17}$$

where the left-hand variable is Gaussian,

$$\text{Re}\{(\mathbf{x} - \bar{\mathbf{x}})^H \mathbf{w}_n\} \sim \mathcal{N}(0, N_0/2 \parallel \mathbf{x} - \bar{\mathbf{x}} \parallel_2^2) \tag{18}$$

so that

$$\Pr(\mathbf{x} \to \bar{\mathbf{x}}) = Q(1/2\,\lambda_n \parallel \mathbf{x} - \bar{\mathbf{x}} \parallel_2) \tag{19}$$

with

$$\lambda_n = \sqrt{\gamma_n}, \quad \gamma_n = \frac{2|h_n|^2}{N_0} \tag{20}$$

Let $B_n = \tau_n \log_2 M_n$ be the number of bits of U_n in a window, $b_{n,m}(\mathbf{x})$ its m-th bit, and

$$\mathcal{X}_{n,m}(\mathbf{x}) = \{\bar{\mathbf{x}} \in \mathcal{X} : b_{n,m}(\bar{\mathbf{x}}) \neq b_{n,m}(\mathbf{x})\} \tag{21}$$

the set of sequences whose m-th bit differs from that of $\mathbf{x}$. The union bound on the BER of U_n is then

$$P_{B_n} \leq \frac{1}{B_n |\mathcal{X}|} \sum_{m=1}^{B_n} \sum_{\mathbf{x} \in \mathcal{X}} \times \sum_{\bar{\mathbf{x}} \in \mathcal{X}_{n,m}(\mathbf{x})} Q\,(1/2\,\lambda_n \parallel \mathbf{x} - \bar{\mathbf{x}} \parallel_2) \tag{22}$$

The bound is loose at low SNR and tight at moderate and high SNR, as Figs. 4 and 5 show. Its cost grows with $|\mathcal{X}|^2$, so it is practical only for small T_2 and low modulation orders, which is why it is used here for BPSK.

IV. BER ANALYSIS OF THE MR-SIC DETECTOR

A. Far User (U_{2})

Define the amplitude

$$A_{\nu_1,\nu_2} = \nu_1\sqrt{\alpha_1/\kappa_1} + \nu_2\sqrt{\alpha_2/\kappa_2} \tag{23}$$

with κ_n from (3). After the T_2 samples are summed,

$$\mathbf{1}^T\mathbf{r}_2 = h_2\,\mathbf{1}^T\mathbf{x} + \widetilde{w}_2, \quad \widetilde{w}_2 \sim \mathcal{CN}(0, T_2N_0) \tag{24}$$

where the combined signal is

$$\mathbf{1}^T\mathbf{x} = \sqrt{\alpha_2}\,T_2\,x_2 + \sqrt{\alpha_1}\sum_{\ell=1}^{T_2} x_1\,[\ell] \tag{25}$$

The summed noise has variance T_2N_0, so the effective noise-to-signal scaling is captured by

$$\bar{\lambda}_n = \lambda_n/\sqrt{\tau_1} \tag{26}$$

The sum of the τ_1 near-user symbols is not unique: several symbol combinations give the same combined point, and their multiplicities follow Pascal's triangle. For $M_1 = M_2 = 4$, where each rail behaves as BPSK, this gives

$$P_{B_2} = \frac{1}{2^{\tau_1}}\sum_{i=0}^{\tau_1}\binom{\tau_1}{i}\,Q\big(A_{2i-\tau_1,\tau_1}\bar{\lambda}_2\big) \tag{27}$$

As an example, for $\tau_1 = 2$ the combined far-user points have the near-user contribution −2, 0 (twice) and +2 in units of $\sqrt{\alpha_1}$, and (27) becomes ¼[$Q(A_{-2,2}\bar{\lambda}_2)$ + 2$Q(A_{0,2}\bar{\lambda}_2)$ + $Q(A_{2,2}\bar{\lambda}_2)$]. For general square QAM let

$$\delta_i = 2i - \Delta_1\tau_1, \quad \bar{\delta}_m = (2m-1)\tau_1 \tag{28}$$

and let c_i be the generalized binomial coefficients defined by

$$\sum_{i=0}^{\tau_1\Delta_1} c_i\; z^i = \Big(\sum_{j=0}^{\Delta_1} z^j\Big)^{\tau_1} \tag{29}$$

together with the Gray-mapping weights of the k-th bit of U_2, as in the BER of square QAM [29], [30],

$$c_2(m,k) = 2^{k-1} - \left\lfloor \frac{(m-1)2^{k-1}}{\sqrt{M_2}} + \frac{1}{2} \right\rfloor \tag{30}$$

$$c_3(m,k) = (-1)^{\lfloor (m-1)2^{k-1}/\sqrt{M_2} \rfloor} \tag{31}$$

and the normalization

$$\beta = 1/2\sqrt{M_1^{\tau_1}M_2} \tag{32}$$

The BER of the k-th bit is

$$P_{B_2}^{(k)} = \frac{1}{\beta}\sum_{m=1}^{(1-2^{-k})\sqrt{M_2}}\;\sum_{i=0}^{\tau_1\Delta_1} c_i\; c_2(m,k)\,c_3(m,k) \times Q\big(A_{\delta_i,\bar{\delta}_m}\bar{\lambda}_2\big) \tag{33}$$

The dominant term corresponds to the smallest argument,

$$A_{\min} = A_{-\Delta_1\tau_1,\tau_1} = \tau_1\big(\sqrt{\alpha_2/\kappa_2} - \Delta_1\sqrt{\alpha_1/\kappa_1}\big) \tag{34}$$

which gives

$$P_{B_2}^{(k)} \approx \frac{2^k}{\sqrt{M_1^{\tau_1}M_2}}\,Q\big(A_{\min}\bar{\lambda}_2\big) \tag{35}$$

so the far-user BER improves with τ_1 because the prefactor decreases and the argument of $Q(\cdot)$ increases. Averaging over the bits gives

$$P_{B_2} = \frac{1}{\log_2\sqrt{M_2}}\sum_{k=1}^{\log_2\sqrt{M_2}} P_{B_2}^{(k)} \tag{36}$$

B. Near User (U_{1})

U_1 first detects x_2, cancels it, and then detects its own symbol, so its BER depends on the SIC outcome. Let $\mathcal{A}$ denote an error in x_1, $\mathcal{B}_C$ a correct and $\mathcal{B}_I$ an incorrect SIC decision. Then

$$P_{B_1} = \Pr(\mathcal{B}_C, \mathcal{A}) + \Pr(\mathcal{B}_I, \mathcal{A}) \tag{37}$$

The noise terms entering $\mathcal{A}$ and $\mathcal{B}$ are jointly Gaussian with correlation coefficient $\rho = 1/\sqrt{T_2}$, which is weak, so the two events are treated as independent:

$$\rho = \frac{\mathrm{Cov}(y_1,y_2)}{\sigma_{y_1}\sigma_{y_2}} = \frac{1}{\sqrt{T_2}}, \quad \Pr(\mathcal{A},\mathcal{B}) \approx \Pr(\mathcal{A})\Pr(\mathcal{B}) \tag{38}$$

For a given transmitted sequence, with $\Theta_{|C} = \Pr(\mathcal{A}|\mathcal{B}_C)$, $\varepsilon_{|C} = \Pr(\mathcal{A}|\mathcal{B}_I)$ and $\xi_{|C} = \Pr(\mathcal{B}_I)$,

$$P_{B_1}^{(i)} = \Theta_{|C_i}\big(1 - \xi_{|C_i}\big) + \varepsilon_{|C_i}\,\xi_{|C_i} \tag{39}$$

Because an incorrect SIC decision can take $M_2 - 1$ values, the explicit expression is derived for BPSK on both users. Let P_{B1}^{lb} denote the BER of U_1 with perfect SIC (given in (45)). Then

$$P_{B_1}^{\mathrm{lb}}(\alpha_1) \le P_{B_1} \le P_{B_1}|_{T_2=1} \tag{40}$$

where the upper bound is the SR-NOMA case. The BER has the general form

$$P_{B_1} = \sum_i c_i\; F_i(\mathrm{Arg}_i), \quad F_i \in \{Q(a\lambda_1),\; Q(a\lambda_1)Q\big(b\bar{\lambda}_1\big)\} \tag{41}$$

with five groups of terms. Writing $s_n = \sqrt{\alpha_n}$ and $a_j = (2j - T_2)s_1 + T_2s_2$, the BPSK case is

$$\begin{aligned} P_{B_1} = \;& Q(s_1\lambda_1) + 2^{-T_2}\{\textstyle\sum_{j=0}^{T_2-1}\binom{T_2-1}{j}[Q\big(a_j\bar{\lambda}_1\big) \\ & + Q\big((s_1+2s_2)\lambda_1\big)Q\big(a_{j+1}\bar{\lambda}_1\big) \\ & - Q\big((2s_2-s_1)\lambda_1\big)Q\big(a_j\bar{\lambda}_1\big)] \\ & - \textstyle\sum_{j=0}^{T_2}\binom{T_2}{j}\,Q(s_1\lambda_1)Q\big(a_j\bar{\lambda}_1\big)\} \end{aligned} \tag{42}$$

The products of two Q-functions are small at moderate and high SNR, and the BER is dominated by the term with the smallest argument, $P_{B1} \approx 2^{-\tau_1}\,Q(\tau_1(\sqrt{\alpha_2} - \sqrt{\alpha_1})\bar{\lambda}_1)$. Table IV shows that (42) agrees with simulation to within simulation accuracy over the whole SNR range.

V. POWER ALLOCATION FOR MR-NOMA

In MR-NOMA the BER depends on both α_1 and T_2. For given $\{\gamma_n, M_n, T_n\}$ and a BER threshold ζ, two problems are considered. The first minimizes the near-user BER subject to a far-user target:

$$\begin{aligned} \min_{\alpha_1} \quad & P_{B_1}(\alpha_1) \\ \text{s.t.} \quad & P_{B_2}(\alpha_1) \le \zeta,\; \alpha_1 < \alpha_{1,\max} \end{aligned} \tag{43}$$

The second minimizes the far-user BER subject to a near-user target:

$$\begin{aligned} \min_{\alpha_1} \quad & P_{B_2}(\alpha_1) \\ \text{s.t.} \quad & P_{B_1}(\alpha_1) \le \zeta,\; \alpha_1 < \alpha_{1,\max} \end{aligned} \tag{44}$$

Here P_{B2} increases and P_{B1} decreases with α_1 over the admissible range, so each optimum lies on a constraint boundary. The near-user BER of MR-NOMA has no tractable

closed form, so, following [26], it is replaced by its perfect-SIC lower bound, i.e. the single-user BER of M_1-QAM with power fraction α_1 [29],

$$P_{B_1}^{\mathrm{lb}}(\alpha_1) = \frac{2}{\sqrt{M_1}\log_2\sqrt{M_1}} \sum_{k=1}^{\log_2\sqrt{M_1}} \sum_{m=1}^{(1-2^{-k})\sqrt{M_1}} c_2\, c_3\, Q\left((2m-1)\sqrt{\alpha_1/\kappa_1}\ \lambda_1\right) \qquad (45)$$

where c_2 and c_3 are given by (30) and (31) with M_2 replaced by M_1. This bound does not depend on T_2. For problem (43) the solution is

$$\alpha_1^* = \min\{\alpha_{1,\max},\ \tilde{\alpha}\}, \quad P_{B_2}(\tilde{\alpha}) = \zeta \qquad (46)$$

and for problem (44) it satisfies

$$\alpha_1^* = \hat{\alpha}, \quad P_{B_1}^{\mathrm{lb}}(\hat{\alpha}) = \zeta \qquad (47)$$

and is therefore independent of T_2. Both problems are solved numerically on a uniform grid of 2000 values of α_1 in $[10^{-4}, \alpha_{1,\max} - 10^{-4}]$, using (36) for P_{B2}. Since longer symbols reduce the rate of the far user, the sum rate in bits per sample,

$$R_{\mathrm{sum}} = \sum_{n=1}^{N} \frac{\log_2 M_n}{T_n} \text{ bits per sample} \qquad (48)$$

is reported together with the optimal allocations to expose the trade-off between rate and reliability.

VI. NUMERICAL RESULTS

A. Setup

All curves are computed from the analytical expressions (22), (36), (42) and (45) with the parameters in Table III. For [2,2], where the closed-form near-user expression applies, the results are additionally checked by Monte Carlo simulation of both detectors (Appendix A).

TABLE III
Simulation parameters

Parameter	Value
Channel	AWGN, $h_n = 1$, SNR = $1/N_0$
Users / durations	$N = 2$, $[T_1, T_2] = [1, T_2]$, $T_2 \in \{1, 2, 4, 8\}$
Modulation pairs $[M_1,M_2]$	[2,2], [4,4], [16,16], [64,64], [4,16], [4,64], [16,4], [16,64], [64,4], [64,16]
Power allocation	Equal distance (5), unless stated
Optimization	$\zeta = 10^{-2}$, $M_2 = 4$, $\gamma_1 = 16, 19, 28$ dB for $M_1 = 4, 16, 64$; $\gamma_2 = 8$ dB

B. JMLSD Versus MR-SIC, and Validation

Figs. 4 and 5 compare the two detectors for BPSK on both users and $T_2 = 2$, together with simulation. In Fig. 4, $[\alpha_1, \alpha_2] = [0.2, 0.8]$. The JMLSD union bound lies above the MR-SIC curve at low SNR; at 5 dB it gives 0.20 for U_1 against 0.143 for MR-SIC, and 2.1×10^{-2} for U_2 against 1.4×10^{-2}. From about 10 dB onward the curves coincide, with $P_{B2} \approx 5.9\times10^{-4}$ at 10 dB and $P_{B1} \approx 1.9\times10^{-4}$ at 15 dB. The simulated BER of the JMLSD receiver, run at 0, 5, 10, 12 and 14 dB, is identical to that of MR-SIC, as expected from their equivalence, and it follows the closed-form MR-SIC curves (Table IV). The gap to the JMLSD curves at low SNR therefore reflects only the looseness of the union bound.

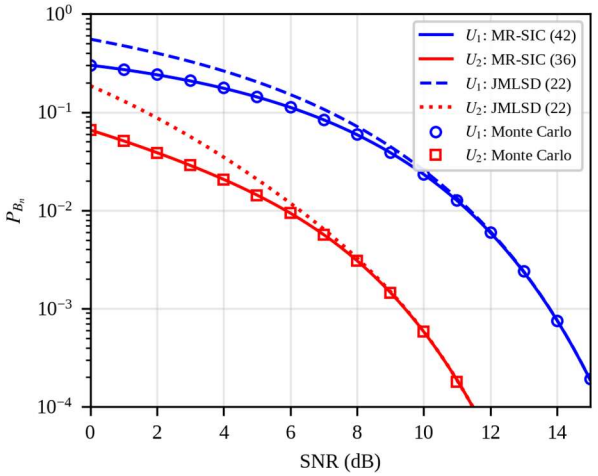


Fig. 4. BER of JMLSD (union bound, (22)) and MR-SIC ((36), (42)) for [2,2] and $T_2 = 2$ versus SNR with $[\alpha_1, \alpha_2] = [0.2, 0.8]$. Markers: Monte Carlo simulation.

TABLE IV
Validation for [2,2], $T_2 = 2$, $[\alpha_1, \alpha_2] = [0.2, 0.8]$

SNR (dB)	P_{B1} sim.	P_{B1} (42)	P_{B1} (22)	P_{B2} sim.	P_{B2} (36)	P_{B2} (22)
0	3.01×10^{-1}	3.00×10^{-1}	5.53×10^{-1}	6.56×10^{-2}	6.57×10^{-2}	1.85×10^{-1}
5	1.44×10^{-1}	1.43×10^{-1}	2.03×10^{-1}	1.43×10^{-2}	1.43×10^{-2}	2.08×10^{-2}
10	2.34×10^{-2}	2.33×10^{-2}	2.57×10^{-2}	5.88×10^{-4}	5.85×10^{-4}	5.89×10^{-4}
12	5.98×10^{-3}	5.95×10^{-3}	6.13×10^{-3}	4.95×10^{-5}	4.62×10^{-5}	4.62×10^{-5}
14	7.75×10^{-4}	7.64×10^{-4}	7.67×10^{-4}	7.50×10^{-7}	9.21×10^{-7}	9.21×10^{-7}

Simulation: 4×10^6 windows per SNR point; (22) is the JMLSD union bound.

Fig. 5 shows the BER versus α_1 at 15 dB. The far-user BER increases monotonically with α_1, whereas the near-user BER is convex: it falls as U_1 receives more power, reaches a minimum of 3.5×10^{-5} at $\alpha_1 \approx 0.26$, and then rises because SIC errors begin to dominate. For $\alpha_1 \gtrsim 0.3$ the two users have almost the same BER (1.5×10^{-4} at $\alpha_1 = 0.3$), showing that U_1 errors are caused mostly by wrong SIC decisions; there the two detectors give identical curves. The optimum of the near user alone is thus found well inside the admissible range, which motivates the constrained formulation of Section V.

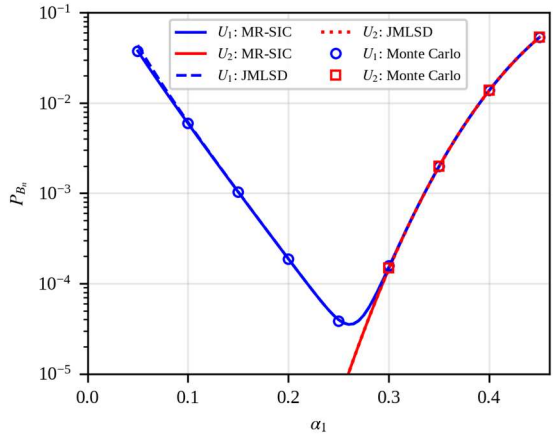


Fig. 5. BER versus α_1 at SNR = 15 dB for [2,2] and $T_2 = 2$. Markers: Monte Carlo simulation.

C. Far-User BER Versus SNR

Figs. 6–10 show the far-user BER versus SNR for the equal-distance allocation and the ten modulation pairs. Table V lists the SNR needed for $P_{B2} = 10^{-2}$. Without rate diversity ($T_2 = 1$), QPSK needs 3.0 dB more than BPSK (13.2 versus 10.2 dB), and 16-QAM and 64-QAM need 13.2 and 21.9 dB more, in line with the values reported in [26].

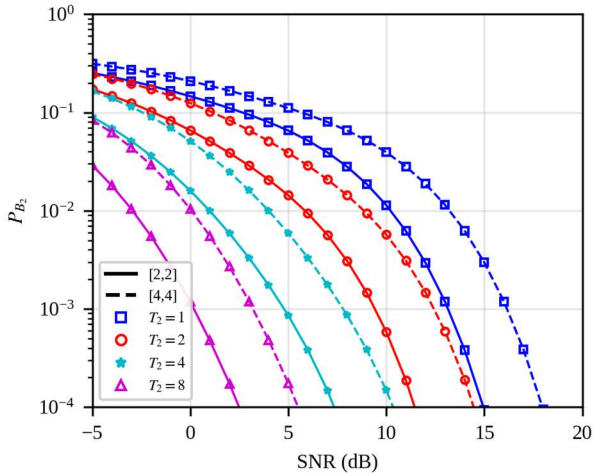


Fig. 6. Far-user MR-SIC BER versus SNR for [2,2] (solid) and [4,4] (dashed) and $T_2 \in \{1, 2, 4, 8\}$.

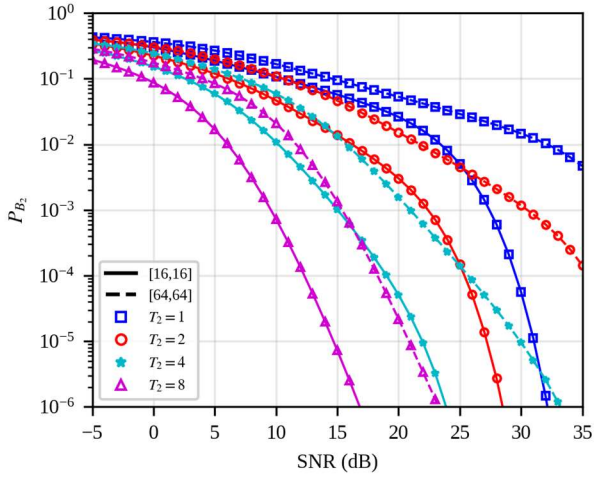


Fig. 7. Far-user MR-SIC BER versus SNR for [16,16] (solid) and [64,64] (dashed) and $T_2 \in \{1, 2, 4, 8\}$.

TABLE V
SNR (dB) required for $P_{B2} = 10^{-2}$

$[M_1,M_2]$	T_2=1	T_2=2	T_2=4	T_2=8	Gain
[2,2]	10.2	5.9	1.0	-2.9	13.2
[4,4]	13.2	8.9	4.0	0.1	13.2
[16,16]	23.4	16.2	10.2	6.1	17.4
[64,64]	32.1	21.8	15.7	11.7	20.5
[4,16]	18.9	14.4	9.6	5.8	13.1
[4,64]	24.5	19.8	15.2	11.4	13.1
[16,4]	18.1	11.1	4.8	0.5	17.6
[16,64]	28.8	21.3	15.6	11.6	17.2
[64,4]	22.2	11.9	5.0	0.6	21.6
[64,16]	27.2	16.8	10.3	6.1	21.1

Gain = SNR(T_2=1) − SNR(T_2=8).

Each doubling of T_2 gives a 3 dB processing gain because the combined energy doubles. On top of this, for even T_2 the near-user interference partly cancels in the sum, which gives the weak-interference cancellation gain G_{WIC}. The total gain over SR-NOMA decomposes as

$$G(T_2) = \mathrm{SNR}_{T_2=1} - \mathrm{SNR}_{T_2} = 10\log_{10} T_2 + G_{\mathrm{WIC}}(T_2) \quad (49)$$

For BPSK/QPSK the total gain at $T_2 = 8$ is 13.2 dB, i.e. 9 dB plus 4.2 dB; for [16,16] and [64,64] it is 17.4 and 20.5 dB, so G_{WIC} is 8.4 and 11.5 dB. The additional gain therefore grows with the near-user modulation order: with $M_2 = 4$ fixed, the total gain at $T_2 = 8$ is 13.2, 17.6 and 21.6 dB for $M_1 = 4$, 16 and 64 (Table V). A consequence is that long symbols make the far user almost insensitive to the near-user modulation: at $T_2 = 1$, raising M_1 from 4 to 64 costs the far user 9.0 dB (13.2 to 22.2 dB), whereas at $T_2 = 8$ it costs only 0.5 dB (0.1 to 0.6 dB).

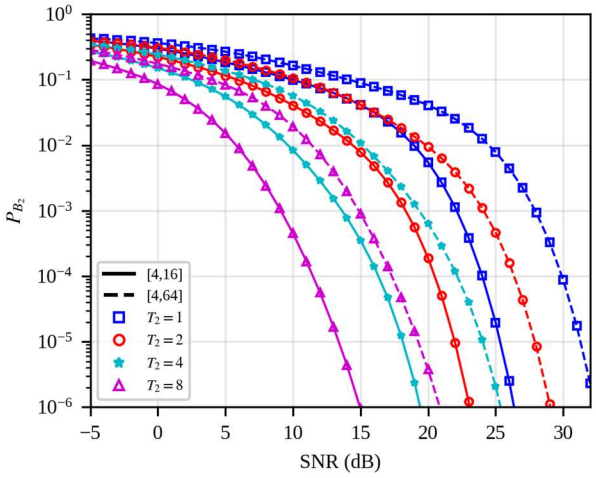


Fig. 8. Far-user MR-SIC BER versus SNR for [4,16] (solid) and [4,64] (dashed) and $T_2 \in \{1, 2, 4, 8\}$.

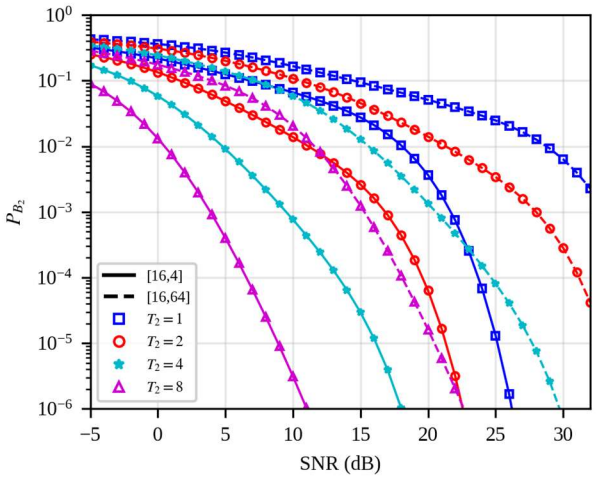


Fig. 9. Far-user MR-SIC BER versus SNR for [16,4] (solid) and [16,64] (dashed) and $T_2 \in \{1, 2, 4, 8\}$.

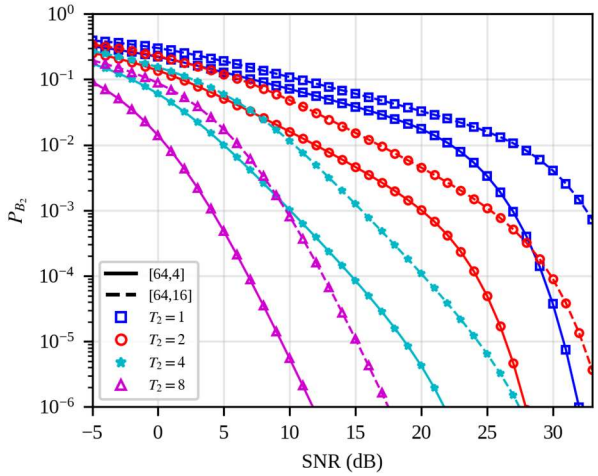


Fig. 10. Far-user MR-SIC BER versus SNR for [64,4] (solid) and [64,16] (dashed) and $T_2 \in \{1, 2, 4, 8\}$.

The order of the far-user modulation matters in the usual way: at $T_2 = 1$ changing M_2 from 16 to 64 for $M_1 = 4$ costs 5.6 dB (18.9 to 24.5 dB), for $M_1 = 16$ from 4 to 64 it costs 10.7 dB, and for $M_1 = 64$ from 4 to 16 it costs 5.0 dB. The symbol duration also provides finer rate steps. At a far-user rate of 1 bit per T_1, QPSK with $T_2 = 2$ needs 8.9 dB, which is 1.3 dB less than BPSK with $T_2 = 1$ (10.2 dB); at 0.5 bit, QPSK with $T_2 = 4$ needs 4.0 dB against 5.9 dB for BPSK with $T_2 = 2$, a gain of 1.9 dB. The comparison is not always favorable: at 2 bits, [16,16] with $T_2 = 2$ (16.2 dB) is 3.0 dB worse than [4,4] with $T_2 = 1$ (13.2 dB).

D. Detection Complexity

Table II shows that the JMLSD cost grows exponentially with T_2 since $|\mathcal{X}| = 4^{T_2+1}$, whereas the MR-SIC cost grows linearly as $39T_2 + 37$ multiplications. This allows the use of long symbol durations, which are the ones that give the largest gains, without exhaustive sequence search.

E. Optimal Power Allocation

Figs. 11 and 12 and Table VI give the solutions of (43) and (44) for $\zeta = 10^{-2}$, $[M_1, 4]$ with $M_1 = 4, 16, 64$, $\gamma_1 = 16, 19, 28$ dB, and $\gamma_2 = 8$ dB. The near-user BER is evaluated with the lower bound (45).

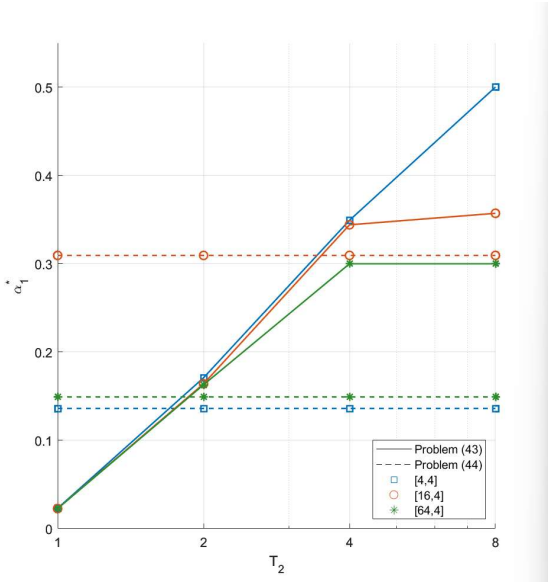


Fig. 11. Optimal power allocation α_1^* versus T_2 for problems (43) (solid) and (44) (dashed). Markers: □ [4,4], ○ [16,4], ✳ [64,4].

For problem (43), α_1^* increases with T_2 and then saturates at $\alpha_{1,\max}$. At $T_2 = 1$ (SR-NOMA) the far-user target allows only $\alpha_1^* \approx 0.023$ for all three pairs. At $T_2 = 2$ it rises to 0.171, 0.164 and 0.163, at $T_2 = 4$ to 0.349, 0.344 and 0.300, and at $T_2 = 8$ to 0.500, 0.357 and 0.300 for [4,4], [16,4] and [64,4]. The [64,4] pair reaches its limit $\alpha_{1,\max} = 0.3$ already at $T_2 = 4$ and [16,4] reaches 0.357 at $T_2 = 8$. The explanation is that a larger T_2 reduces P_{B2} for the same α_1, so the far user can meet ζ with less power and the remainder can be given to U_1. For problem (44), $\alpha_1^* = 0.136$, 0.309 and 0.149 and does not change with T_2, because the lower bound (45) does not depend on T_2. The values differ between pairs because α_1^* depends on both M_1 and γ_1.

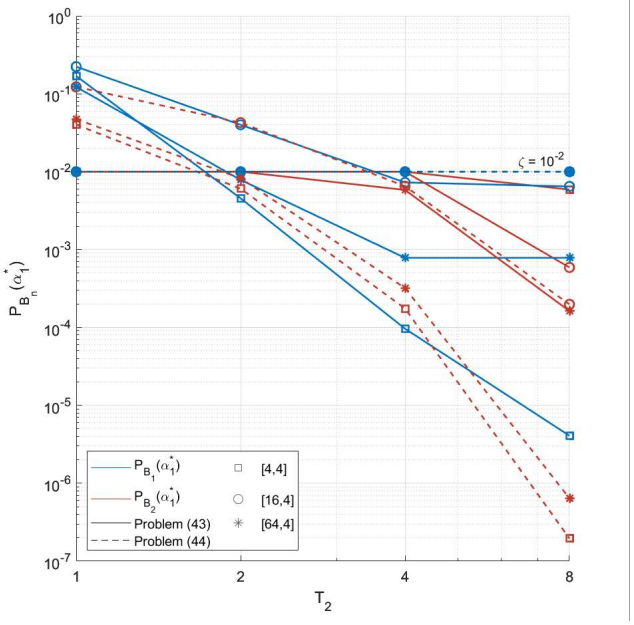


Fig. 12. BER at the optimal allocation, $P_{B1}(\alpha_1^*)$ and $P_{B2}(\alpha_1^*)$, versus T_2 for problems (43) (solid) and (44) (dashed). Markers: □ [4,4], ○ [16,4], ✳ [64,4].

Fig. 12 shows the resulting BERs. For problem (43), $P_{B2}(\alpha_1^*)$ equals ζ as long as $\alpha_1^* < \alpha_{1,\max}$, and falls below it afterwards (5.9×10^{-3} for [4,4] and 5.9×10^{-4} for [16,4] at $T_2 = 8$). The near-user BER keeps improving with T_2: for [4,4] it falls from 0.17 at $T_2 = 1$ to 4.1×10^{-6} at $T_2 = 8$, and for [64,4] from 0.12 to 7.8×10^{-4}, where it saturates together with α_1^*. For problem (44), $P_{B1}(\alpha_1^*) = \zeta$ for all T_2, and $P_{B2}(\alpha_1^*)$ decreases with T_2; for [4,4] it goes from 4.0×10^{-2} at $T_2 = 1$ to 6.0×10^{-3} at $T_2 = 2$ and 1.9×10^{-7} at $T_2 = 8$. Since P_{B1} is replaced by its lower bound, the true near-user BER can be higher when α_1^* is close to $\alpha_{1,\max}$.

TABLE VI
Optimal power allocations for problems (43) and (44)

$[M_1,M_2]$	T_2	α_1^* (43)	P_{B1} (43)	P_{B2} (43)	α_1^* (44)	P_{B2} (44)
[4,4]	1	0.023	0.17	1.0×10^{-2}	0.136	4.0×10^{-2}
[4,4]	2	0.171	4.5×10^{-3}	1.0×10^{-2}	0.136	6.0×10^{-3}
[4,4]	4	0.349	9.6×10^{-5}	1.0×10^{-2}	0.136	1.7×10^{-4}
[4,4]	8	0.500	4.1×10^{-6}	5.9×10^{-3}	0.136	1.9×10^{-7}
[16,4]	1	0.023	0.22	1.0×10^{-2}	0.309	0.12
[16,4]	2	0.164	4.0×10^{-2}	1.0×10^{-2}	0.309	4.3×10^{-2}
[16,4]	4	0.344	7.3×10^{-3}	1.0×10^{-2}	0.309	6.5×10^{-3}
[16,4]	8	0.357	6.5×10^{-3}	5.9×10^{-4}	0.309	2.0×10^{-4}
[64,4]	1	0.023	0.12	1.0×10^{-2}	0.149	4.7×10^{-2}
[64,4]	2	0.163	7.9×10^{-3}	1.0×10^{-2}	0.149	8.2×10^{-3}
[64,4]	4	0.300	7.8×10^{-4}	5.8×10^{-3}	0.149	3.2×10^{-4}
[64,4]	8	0.300	7.8×10^{-4}	1.6×10^{-4}	0.149	6.4×10^{-7}

P_{B1} uses the perfect-SIC bound (45); $P_{B1}(\alpha_1^*) = \zeta$ for (44).

F. Rate–Reliability Trade-off

Longer far-user symbols reduce the sum rate (48). For [4,4] it falls from 4 to 3, 2.5 and 2.25 bits per sample as T_2 goes from 1 to 2, 4 and 8, for [16,4] from 6 to 4.25 bits, and for [64,4] from 8 to 6.25 bits. In return, the near-user BER at the optimum of (43) improves by more than four orders of magnitude for [4,4]. The relative rate loss is smaller for higher M_1: from $T_2 = 1$ to 8 the sum rate drops by 44 % for [4,4] but only by 22 % for [64,4]. MR-NOMA is therefore most attractive when a high-order near user is paired with a low-rate far user, since the far-user energy gain also stabilizes the SIC stage.

G. Discussion

Three observations are useful for system design. First, because a long far-user symbol makes the combined far-user signal nearly independent of the near-user constellation, the near user can use a high-order QAM without a large penalty for the far user. Second, problem (44) gives a T_2-independent allocation under the lower bound, so it can be computed offline, whereas the solution of (43) must be recomputed for every T_2 until it saturates. Third, the results assume AWGN, perfect knowledge of the channel at the receivers, and two users; fading, channel estimation errors and larger groups will reduce the gains and are left for future work.

VII. CONCLUSION

We have implemented, validated by simulation and evaluated the analytical BER of downlink two-user MR-NOMA. The MR-SIC detector matches the optimal JMLSD at moderate and high SNR while replacing a search over $M_1^{T_2}M_2$ sequences by a few single-user decisions. Longer far-user symbols give a 3 dB gain per doubling plus a weak-interference cancellation gain that increases with the near-user modulation order, reaching 21.6 dB at $T_2 = 8$ for [64,4]. In the BER-constrained power allocation, T_2 relaxes the far-user constraint, so that the near-user power share rises until the reliable-SIC limit $\alpha_{1,\max}$ is reached, at the cost of a lower sum rate. Future work includes fading channels, uplink MR-NOMA, more than two users and simulation of the higher-order QAM cases.

APPENDIX A: SIMULATION PROCEDURE

The Monte Carlo results follow the procedure of Algorithm 1. For each SNR point, K windows of T_2 samples are generated: the T_2 near-user BPSK symbols and the far-user symbol are drawn uniformly, superposed with the coefficients $[\alpha_1, \alpha_2]$, and complex Gaussian noise of variance $N_0 = 1/\mathrm{SNR}$ is added. The two receivers then operate on the same noise realization.

Step 1: U_2 sums the T_2 received samples and decides $\hat{x}_2$ by the sign of the real part.

Step 2: U_1 subtracts $\sqrt{\alpha_2}\hat{x}_2$ from every sample and decides each near-user symbol from the sign of the real part of its own sample.

Step 3 (JMLSD): all 2^{T_2+1} candidate sequences are tested with the metric (7), and the closest to the received window is selected.

Step 4: bit errors are counted over K T_2 near-user bits and K far-user bits.

We used $K = 3\times10^6$ per point for Fig. 4, $K = 4\times10^6$ for Table IV and $K = 6\times10^6$ for Fig. 5 (MR-SIC only). Points with no observed error are not plotted.

APPENDIX B: NUMERICAL ASPECTS

The far-user expression (33) is an alternating-sign sum whose coefficients, obtained as the T_2-fold convolution of a vector of ones in (29), grow quickly with T_2 and M_1. In double precision the cancellation limits the reliable range of the result to roughly 10^{-9}, so values below this level should not be interpreted. For the SNR range used here the lowest BER shown is 10^{-6}. The perfect-SIC bound (45) does not have this issue. The coefficients c_i are the entries of a convolution power, which also equals the number of ways in which the near-user symbols can add to the same combined point.

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